

Deploying a Machine Learning Model

Project: Iris Flower Classification API using Flask

Abstract

This project demonstrates how to deploy a trained machine learning model as a web API. An Iris flower classification model is trained using a Random Forest classifier and exposed through a Flask application. The API accepts flower measurements and returns the predicted Iris species.

Problem Statement

Develop a machine learning application that predicts the Iris flower species and make the model available through a REST API for external applications.

Objectives

Train a classification model; save the trained model; create a prediction API using Flask; prepare the application for deployment; verify predictions using API requests.

Dataset

The Iris dataset contains 150 samples with four numerical features: sepal length, sepal width, petal length, and petal width. The target classes are Setosa, Versicolor, and Virginica.

Tools and Technologies

Python, Flask, Scikit-learn, NumPy, Pickle, Jupyter Notebook.

Methodology

Load the dataset, preprocess if required, train a Random Forest classifier, evaluate accuracy, save the trained model as model.pkl, build a Flask API (app.py), and expose a /predict endpoint.

API Workflow

Client sends a POST request containing four flower measurements. The Flask application loads model.pkl, predicts the class, and returns the predicted species in JSON format.

Expected Results

The API successfully classifies Iris flowers and returns predictions with low response time. The trained model can be reused without retraining for each request.

Advantages

Simple deployment workflow, reusable model, easy API integration, scalable for future cloud deployment.

Future Scope

Containerize the application using Docker, deploy to AWS, Google Cloud, or Azure, add authentication, logging, monitoring, and CI/CD pipelines.

Conclusion

This project demonstrates the complete workflow from machine learning model training to API deployment readiness. It provides a foundation for deploying larger machine learning applications in production.

References

1. Scikit-learn Documentation
2. Flask Documentation
3. Iris Dataset Documentation